ChargeTrust

Complete Community EV Charging Plan

Tagline: Find with confidence. Charge with proof. Pay fairly.

1. Core Problem

- EV consumers without home charging struggle to find a nearby, compatible, genuinely available and affordable charger.
- Apartment residents face permission problems, shared parking, limited electrical capacity, queues and difficult electricity billing.
- Hosts with unused chargers worry about unknown users, privacy, safety, cancellations, no-shows and unclear revenue sharing.
- Existing platforms commonly focus on listing, discovery, booking and payment; the deeper gap is proving that the actual charging transaction happened successfully.

2. Target Users

- Primary: renters and apartment residents who cannot install private chargers.
- Secondary: home hosts, shops, offices, apartment societies, taxi drivers, delivery drivers and EV two-wheeler users.

3. Core Solution

Build a community EV-charging app connecting consumers who need charging with hosts and societies that provide charging. The app manages discovery, verification, reservation, check-in, charging proof, metering, billing, settlement and failure protection.

4. Main Innovation: Charging Proof Chain

Verified host → Verified charger → Verified parking → Verified consumer arrival → Verified charging start → Recorded energy consumption → Verified completion → Fair payment settlement

5. Charging Passport

Every charger receives a digital profile that helps consumers decide whether it is genuine, compatible, available, reliable and worth booking.

- Host identity and mobile verification.
- Location and parking verification.
- Charger type, connector, power and operating hours.
- Price, billing method and cancellation policy.
- Current status: available, in use, reserved, offline or under maintenance.
- Successful-session percentage, failure history, ratings and reliability score.
- Safety checklist and maintenance history.

6. User Roles

Consumer

- Search and filter chargers.
- Check Charging Passport.
- Estimate cost and time.

- Reserve a slot and pay a deposit.
- Check in using QR.
- Track the session and receive an invoice.
- Rate the host and charger.

Host

- Register charger and parking.
- Verify identity and location.
- Set price and availability.
- Approve bookings.
- Monitor sessions.
- Receive automatic payment.
- Report maintenance issues.

Society Administrator

- Manage shared chargers and parking.
- Set building load limits.
- Manage residents, visitors and queues.
- Allocate power fairly.
- Generate bills and revenue reports.

7. Complete Consumer Workflow

- **1.** Open ChargeTrust and log in or sign up.
- **2.** Verify mobile number using OTP.
- **3.** Select Consumer role.
- **4.** Add vehicle, connector, battery capacity and current battery level.
- **5.** Search current location or destination.
- **6.** Filter chargers by distance, compatibility, speed, price, availability and parking.
- **7.** Open the selected charger's Charging Passport.
- **8.** Check host, charger, parking, safety, reliability and billing details.
- **9.** Enter current and target battery percentage.
- **10.** View estimated energy, duration, electricity cost, parking cost, platform fee and tax.
- **11.** Select a time slot or join a queue.
- **12.** Confirm booking and refundable deposit.
- **13.** Pay using UPI, card or wallet.
- **14.** Receive booking ID, QR code, parking bay and access instructions.
- **15.** Navigate to the charger.
- **16.** Scan the physical charger QR code on arrival.
- **17.** Confirm correct location, parking and charger condition.
- **18.** Host approves access if manual approval is enabled.
- **19.** Connect vehicle and start charging.
- **20.** Track energy consumed, power, duration, cost and expected completion.
- **21.** Stop charging, record final meter reading and release parking.
- **22.** Receive final bill based on actual usage.
- **23.** Pay the remaining amount and receive invoice.
- **24.** Rate host, charger, parking, safety and billing.
- **25.** Update booking history and charger reliability score.

8. Host Workflow

- Register as host and verify identity.
- Add charger, parking, price, operating hours and supported vehicles.
- Upload charger, parking and location photos.
- Complete safety checklist.
- Select smart-meter, telemetry, manual-meter or time-based billing.
- Receive Charging Passport after review.
- Receive and approve booking requests.
- Monitor arrival and charging session.
- Confirm completion and receive earnings.
- View ratings and report maintenance.

9. Society Workflow

- Register society and administrator.
- Add chargers and parking bays.
- Set maximum charging capacity and peak hours.
- Verify residents and visitors.
- Monitor active chargers and total load.
- Prioritize vehicles by departure time and urgency.
- Allocate, reduce, pause or resume charging power.
- Notify consumers of changed completion time.
- Generate resident and visitor bills.
- Display revenue and maintenance reserve.

10. Cost Estimation

Estimated energy: E = (Battery capacity × (Target SOC – Current SOC)) ÷ Charging efficiency

Estimated bill = (Energy required × Energy tariff) + Parking cost + Platform fee + Tax

The app clearly marks this as an estimate. The final bill uses actual meter or session data.

11. Final Billing and Settlement

- Energy consumed and electricity tariff.
- Parking duration and parking fee.
- Platform fee and tax.
- Deposit adjustment, refund or penalty.
- Electricity reimbursement, host earnings, society share, platform fee and maintenance reserve.

12. Failure Handling

- Host cancels: freeze payment, refund consumer, suggest alternative, apply host penalty.
- Parking occupied: upload photo, notify host or society, reassign or refund.
- Charger fails: stop session, freeze payment, refund, suggest alternative and create maintenance ticket.
- Consumer no-show: start grace period, release slot and apply no-show policy.
- Meter dispute: compare readings and timestamps, review evidence, freeze settlement and correct or refund.

13. Technical Architecture

- Frontend: consumer app, host dashboard, society dashboard and admin panel.
- Backend: authentication, roles, chargers, verification, bookings, QR check-in, sessions, meters, billing, payments, refunds, ratings, notifications and maintenance.
- Database: users, vehicles, chargers, parking spaces, verification records, availability, bookings, sessions, meter readings, invoices, payments, refunds, ratings and society limits.
- Prototype hardware: simulated charger telemetry and meter readings; production integration can use smart chargers or separate meters.

14. One-Month Development Plan

Week 1: Research and design

- Interview EV users, apartment residents and potential hosts.
- Finalize target user and workflow.
- Design wireframes, database and verification rules.
- Freeze MVP scope.

Week 2: Core application

- Build authentication, roles, profiles, charger registration, map/list, filters and booking database.

Week 3: Main innovation

- Build Charging Passport, QR check-in, session states, simulated meter, live session, billing, settlement and refund workflow.

Week 4: Testing and presentation

- Add society dashboard, failure scenarios, ratings and maintenance.
- Run user tests, fix bugs, rehearse demo and prepare pitch.

15. MVP Priorities

- Must build: login, profiles, charger discovery, Charging Passport, reservation, QR check-in, session tracking, simulated meter, final billing, payment distribution and refund.
- If time permits: society load management, queue management, notifications, host analytics, maintenance workflow and UPI test payment.
- Do not prioritize: blockchain, cryptocurrency, hardware manufacturing, advanced AI, nationwide deployment or complex fleet management.

16. Live Demo

Ravi is an apartment renter without a home charger. He registers his EV, finds Anita’s verified charger, checks the Charging Passport, estimates cost, reserves a slot, pays a deposit, reaches the location, scans the QR code, starts charging, sees simulated energy increase, receives a transparent final bill, and rates the session.

Failure demo

Simulate host cancellation after arrival: payment is frozen, Ravi receives a refund, an alternative charger is suggested and the host reliability score decreases.

17. Final Pitch

ChargeTrust is a proof-based community EV-charging platform for renters and apartment residents who cannot charge at home. It verifies the host, charger, parking space, consumer arrival, charging session, energy consumption, final bill and payment settlement—making shared charging safe, transparent and dependable.

One-line explanation: ChargeTrust does not just help users find a charger; it proves that the complete charging transaction happened fairly.